GMI German-Malaysian Institute

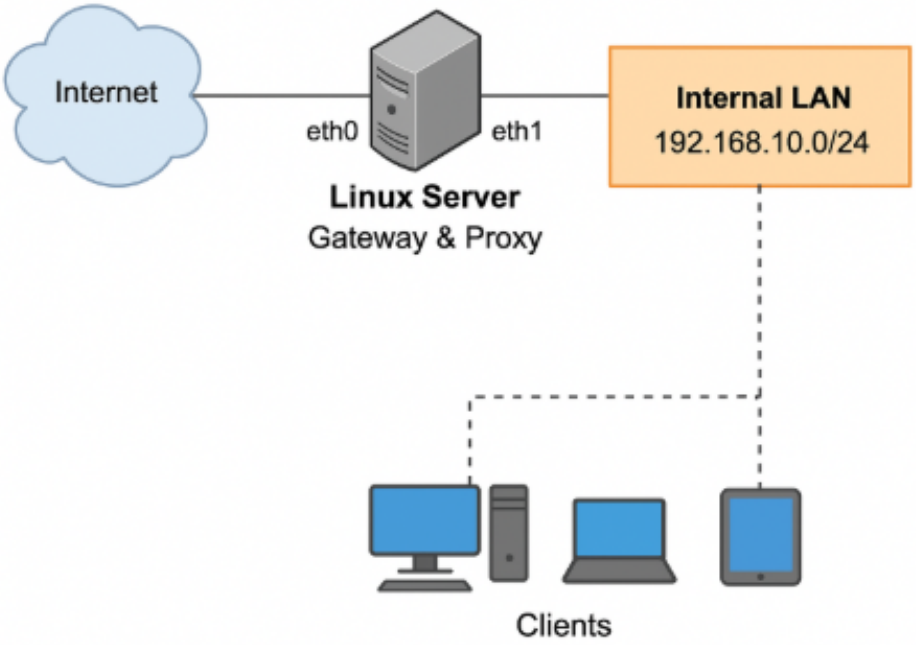
DIPLOMA PROGRAMME

MINI PROJECT 30%

Academic Period: JAN 2026

Code	CBS 2413	Course Name	SECURING NETWORK
Title	WEB PROXY WITH SQUID		
Name	1. Abdul Hariz Mohd bin Sarnudin (CBS24070208		3. Muhammad Danial bin Norhisam (CBS24070222)
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Group	CBS 1	Examiner	MOHD HAFIZ RAZALI

No	Assessment Criteria	CLO	PO	Level of Diff	Full Marks	Score	Marks	Comment
1.	All criteria are based on the rubric provided.	2, 3	2, 6	P3, A3	32			
2								
3.								
4.								
5								
				Total				
				Total Marks	/32			
			Total	PO	/32 X 30%			
				Signature				



QUESTION:

In this project, you are required to design and implement a secure network using iptables and squid proxy that controls and monitors internet access for a small office network. Give the solution includes traffic filtering, access control, and to optimize performance and security.

PROJECT REQUIREMENTS:

Virtualized Network Design:

- Create a virtual network topology within ESXi, simulating a small office or lab environment. This should include at least two LANs representing different network segments – internal and external.
- Document the network design, including IP addressing schemes, LAN configurations, and logical network diagram.

Iptables configuration:

- Allow only HTTP traffic from LAN.
- Block access to specific external websites or IP addresses.

Squid Proxy setup:

- Block access to social media and entertainment websites

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Testing and Verification:

- Develop test cases to verify the functionality of the security policies.
- Document the test results.

Documentation:

- Create a project video, including:
    - Network design documentation.
    - Step-by-step configuration instructions.
    - Test results and analysis.
    - Lessons learned and recommendations for future improvements.
      Include group members at the wallpaper of the video.

Deliverables:

- Virtual network topology diagram.
- Test plan and results.
- Project documentation.

Rules and Regulations:

- Each group may have a maximum of 4 members (permitted to be done individually)
- Each group must complete the project by the assigned deadline within timeline given.
- All group members must actively participate and contribute to the project.
- Ensure that all sources used in the project are properly cited.
- Each group must present their findings clearly (narration and subtitle)
- Late submissions will be penalized according to the project guidelines.

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RUBRICS: SECURING A VIRTUALIZED NETWORK ENVIRONMENT WITH SQUID WEB PROXY

NO.	CRITERIA	EXCELLENT (4)	GOOD (3)	FAIR (2)	NEED IMPROVEMENT (1)	SCORE
	GROUP COMPONENT (TOTAL MARK = 24)					
1	Network Design & Implementation	Flawless design, accurate config.	Minor errors, mostly correct	Noticeable errors, partially correct.	Major errors, incomplete.	/4
2	Squid Proxy Configuration	Perfect deployment, optimized config.	Minor errors, functional.	Noticeable errors, unstable.	Incomplete, non-functional.	/4
3	Security Policy Implementation	Effective, accurate, optimized policies.	Minor errors, functional policies.	Noticeable errors, partial policies.	Incomplete or incorrect policies.	/4
4	Testing & Verification	Comprehensive test plan, thorough testing.	Basic testing, present results.	Incomplete plan, limited testing.	No testing, missing results.	/4
5	Content & Presentation	Clear, well-organized content, professional.	Clear and adequate presentation.	Unclear, poor presentation.	Missing/poor video, no presentation.	/4
6	Group Collaboration	Equal contribution, excellent teamwork.	Mostly good contribution, teamwork.	Uneven contribution, fair teamwork.	Imbalance, poor teamwork.	/4
	INDIVIDUAL COMPONENT (TOTAL GRADE = 8)					
7	Individual Contribution	Detailed, reflective, insightful.	Present, outlines contributions.	Incomplete, vague.	Missing, no description.	/4
8	Individual Technical	Deep understanding, effective troubleshooting.	Good understanding, basic troubleshooting.	Fair understanding, struggles.	Poor understanding, no troubleshooting.	/4
					TOTAL	**/32**

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INTRODUCTION

1.0 PURPOSE OF THE PROJECT

Today, business computer networks face many security risks such as online threats, data leaks, and loss of productivity. Because of this, a secure network gateway is important to protect internal users and control internet access. The main objective of this project is to design and implement a secure network gateway using an Ubuntu Linux Server, iptables, and Squid Proxy.

This project aims to monitor, filter, and control internet traffic between the internal client network and the external internet network. The Ubuntu Server is configured with two network adapters to separate the network into two zones. The external interface connects to the internet using NAT, while the internal interface connects to the private client network using the static IP address 192.168.10.1.

The purpose of this separation is to prevent internal client computers, such as the client using 192.168.10.45, from accessing the internet directly without security control. All client traffic must pass through the Ubuntu Server, where iptables is used for firewall traffic filtering and Squid Proxy is used for website and file type filtering. This project is important because it demonstrates how a small office network can be secured, monitored, and managed using open-source tools.

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2.0 PROJECT BACKGROUND AND STUDY

To understand the design of this project, it is helpful to look at how modern office networks protect themselves. A secure network relies on a strategy called "Defense-in-Depth." This approach uses multiple layers of security so that if one control fails, another layer is there to stop the threat. In this gateway architecture, two different types of security tools are combined on the Ubuntu Server, a network firewall (iptables) and a web filter (Squid Proxy). These tools work together at different levels of the network infrastructure to protect the organizational environment.

I) The Network Layer (iptables)

The first layer of security happens at the network infrastructure level using iptables. Traditional firewalls inspect data packets and filter them based on numerical parameters, such as IP addresses or port numbers, such as Port 80 for web browsing. In this system setup, iptables acts as a strict guard at the boundary. By setting the firewall to drop all unapproved traffic by default, a "zero-trust" environment is established. This means no device inside the internal office network can send data out to the internet unless the firewall explicitly validates and permits the packet stream.

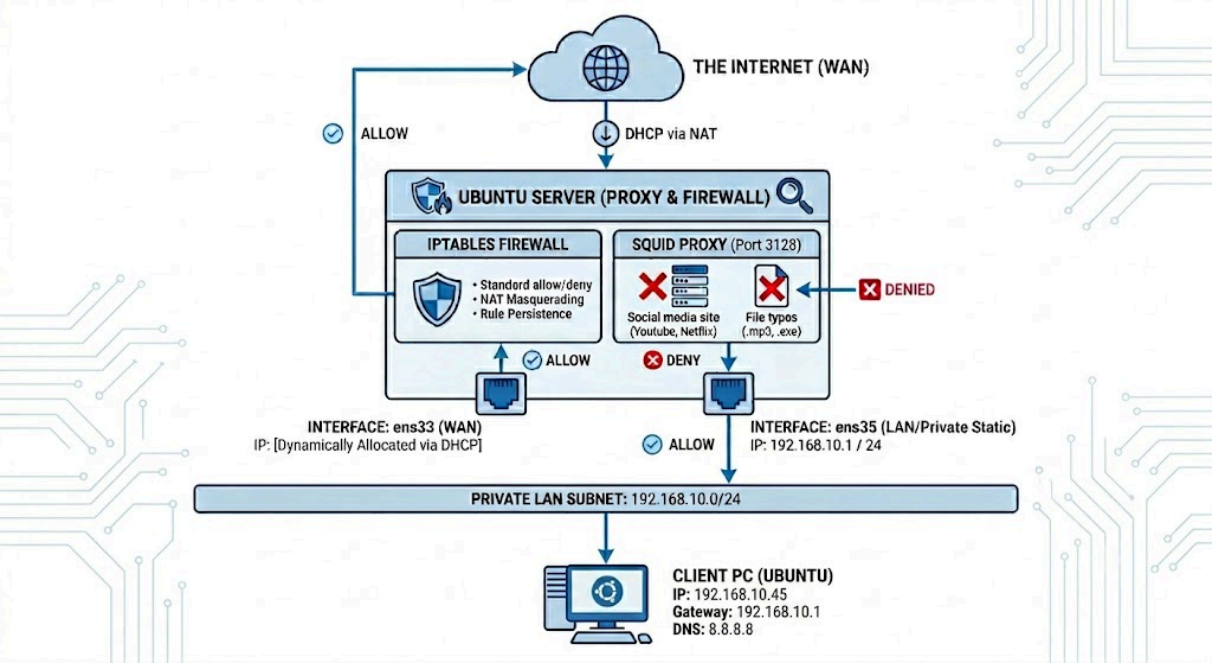
II) The Application Layer (Squid Proxy)

However, basic network firewalls face major limitations when dealing with modern websites like YouTube, Netflix, or TikTok. These large platforms use thousands of dynamically shifting IP addresses across global Content Delivery Networks (CDNs) to deliver their media content. Trying to restrict YouTube by blocking a single IP address is ineffective, as the platform automatically routes traffic through alternative active addresses to bypass the firewall.

This is why Squid Proxy is integrated into the architecture, operating at the application layer (Layer 7). Instead of filtering by random numerical IP records, Squid inspects the actual web domain names (such as .youtube.com or .netflix.com). Regardless of how many times a platform updates its destination IP addresses, Squid recognizes the string domain name and drops the connection instantly. Combining these two distinct security layers provides a complete, secure framework for managing enterprise office network traffic.

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3.0 PROJECT SCOPE AND FOCUS



This mini-project focuses on the design, deployment, and testing of a secure network gateway inside a virtualized environment. The project is restricted to setting up a dual-homed Linux server that controls and filters internet access for a single internal client machine. It does not cover large-scale enterprise deployments, physical hardware routing, or complex Active Directory domain integrations. Instead, the focus is placed on three main operational areas.

I) Virtual Routing and Network Segmentation

1. Interface Isolation: Setting up two separate network profiles (NAT and an isolated Host-Only) on a single Ubuntu Server to create a clear border between the public internet and the private network.
2. IP Routing Allocation: Assigning a fixed static IP address architecture (192.168.10.0/24) to ensure reliable tracking and communication between the gateway and internal devices.
3. Traffic Forwarding: Activating the Linux kernel IP forwarding engine to allow the server to pass web packets across the separate network cards safely.

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II) Network Layer Firewall Control

1. Zero-Trust Rulesets: Creating basic iptables rules that default to dropping all unapproved forwarding traffic.
2. Port Protection: Restricting interface inputs so that only necessary web services, specifically standard HTTP (Port 80) and the Squid communication port (Port 3128), can talk to the server.
3. Network Address Translation (NAT): Applying masquerading rules to mask internal client IP addresses behind the server's public IP when accessing external websites.
4. Rule Persistence: Installing automated tools to guarantee that firewall configurations remain active when the system restarts.

III) Application Layer Web Filtering

1. Domain Restrictions: Configuring Squid Proxy access control lists (ACLs) to target and block specific social media and entertainment platforms (YouTube, Netflix, and Twitch).
2. Policy Sequencing: Arranging proxy access rules in the correct order to ensure explicit block rules run before broad network allowances.
3. Verification Logging: Using system logs (access.log) on the server to track connection requests and verify that blocked attempts generate the correct denial responses.

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4.0 METHODOLOGY

The implementation of this network gateway project relies on a specific set of virtualization and software open-source tools. These tools allow the simulation, configuration, and testing of a real-world enterprise security boundary. Below is a description of the tools used to carry out the project:

I) Hypervisor Platform (VMware Workstation)

- Acts as the Virtualization Host
- Used to create and run the virtual machines for this project. It provides the virtual hardware environments and manages the private network connections. By using its virtual network editor, the systems can be linked together using an isolated Host-Only/LAN network switch, keeping the client machine separated from the outside host computer.

II) Operating Systems (Ubuntu Server & Ubuntu Desktop)

- Acts as the Security Gateway and Client Host
- Ubuntu Server: Functions as the core network gateway. It runs as a lightweight, command-line interface (CLI) system to maximize processing efficiency. It hosts the routing engine, firewall rules, and proxy software.
- Ubuntu Desktop / Client VM: Functions as the local office end-user machine. It provides a Graphical User Interface (GUI) and a web browser to test network connectivity and verify whether security rules are successfully blocking target websites.

III) Linux Networking Engine (Netplan)

- Acts as the Network Interface Configuration
- The default network configuration utility used in modern Ubuntu systems. It reads structured YAML configuration files to assign static IP addresses, define subnet masks, and activate network interfaces (ens33 and ens35). It ensures that the gateway interfaces start up reliably with the correct network parameters upon boot.

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5.0 SIGNIFICANCE OF THE PROJECT

Studying and building a secure network gateway is highly important for several key reasons:

I) Cost-Effective Enterprise Security

For small firms, educational institutions, and nonprofits, proprietary security software and commercial hardware firewalls can be quite costly. This experiment demonstrates that an entirely free, open-source Linux operating system can accomplish the same border security restrictions, offering a cost-effective substitute for IT communities throughout the globe.

II) Mitigating Modern Workplace Risks

Employee productivity is reduced and business internet bandwidth is depleted by unauthorized access to popular video-streaming services like YouTube and Netflix. System administrators may preserve network resources and maintain office data pipelines with the help of this topic's useful, practical solutions.

III) Practical Education and Proficiency Contribution

This project offers a practical guide for network engineers to comprehend how data packets move across various network levels, specifically Layer 3/4 firewalling vs. Layer 7 web filtering. It helps the IT community establish better security practices by bridging the gap between academic classroom concepts and practical system administration abilities.

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6.0 IP ADDRESSING SCHEMES AND LAN CONFIGURATIONS

Device	Network Adapter	Network Type	IP Address	Gateway	Purpose
Server VM / Ubuntu Server	Adapter 1	NAT	DHCP	DHCP	Internet access / external network
Server VM / Ubuntu Server	Adapter 2	Host-only	192.168.10.1/ 24	-	Internal LAN gateway
Client VM	Adapter 1	Host-only	192.168.10.45 /24	192.168.10. 1	Client testing through proxy

Explanation:

The server VM uses two network adapters. The NAT adapter provides internet access, while the Host-only adapter connects the server to the client VM. The client VM is also connected to the Host-only network and uses 192.168.10.1 as its gateway. This setup allows the Ubuntu Server to act as the gateway and Squid Proxy for the internal client network.

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CONTENTS

7.0 NETWORK DESIGN AND IMPLEMENTATION

I) Network Interface Configuration

1. Netplan YAML Configuration File (00-installer-config.yaml)

This configuration file sets up the network adapters on the Ubuntu Server. The external adapter ens33 is configured via DHCP for internet routing, while the internal gateway adapter ens35 is assigned a static IP of 192.168.10.1/24 to establish the private office LAN segment.

```
ubnt4@vm:~$ sudo nano /etc/netplan/00-installer-config.yaml
```

```
  GNU nano 4.8                  /etc/netplan/00-installer-config.yaml
# This is the network config written by 'subiquity'
network:
  version: 2
  renderer: networkd
  ethernets:
    ens33:
      dhcp4: yes
      optional: true

    ens35:
      dhcp4: no
      addresses: [192.168.10.1/24]
```

2. Validation of Netplan Apply and Active IP Address Schemes

Verification screen showing successful execution of sudo netplan applyconf. Running ip addr confirms both network adapters are up, with ens35 bound to 192.168.10.1.

```
ubnt4@vm:~$ sudo netplan generate
ubnt4@vm:~$ sudo netplan apply
ubnt4@vm:~$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
       valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:d0:f4:80 brd ff:ff:ff:ff:ff:ff
    inet 192.168.64.160/24 brd 192.168.64.255 scope global dynamic ens33
       valid_lft 1783sec preferred_lft 1783sec
    inet6 fe80::20c:29ff:fed0:f480/64 scope link
       valid_lft forever preferred_lft forever
3: ens35: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:d0:f4:8a brd ff:ff:ff:ff:ff:ff
    inet 192.168.10.1/24 brd 192.168.10.255 scope global ens35
       valid_lft forever preferred_lft forever
    inet6 fe80::20c:29ff:fed0:f48a/64 scope link
       valid_lft forever preferred_lft forever
```


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3. Verification of Linux Kernel IPv4 Forwarding Setup

Terminal outputs demonstrating the activation of system IP forwarding. Modifying the sysctl runtime parameters allows the Ubuntu Server to pass network traffic across interfaces as a router.

```
ubnt4@vm:~$ sudo sysctl -w net.ipv4.ip_forward=1
[sudo] password for ubnt4:
net.ipv4.ip_forward = 1
ubnt4@vm:~$ sudo bash -c 'echo "net.ipv4.ip_forward=1" >> /etc/sysctl.conf'
ubnt4@vm:~$ sudo sysctl -p
```

4. Successful Deployment of Squid Proxy Package Daemon

Terminal output logs confirming the standard update and clean installation execution of the Squid web proxy service environment.

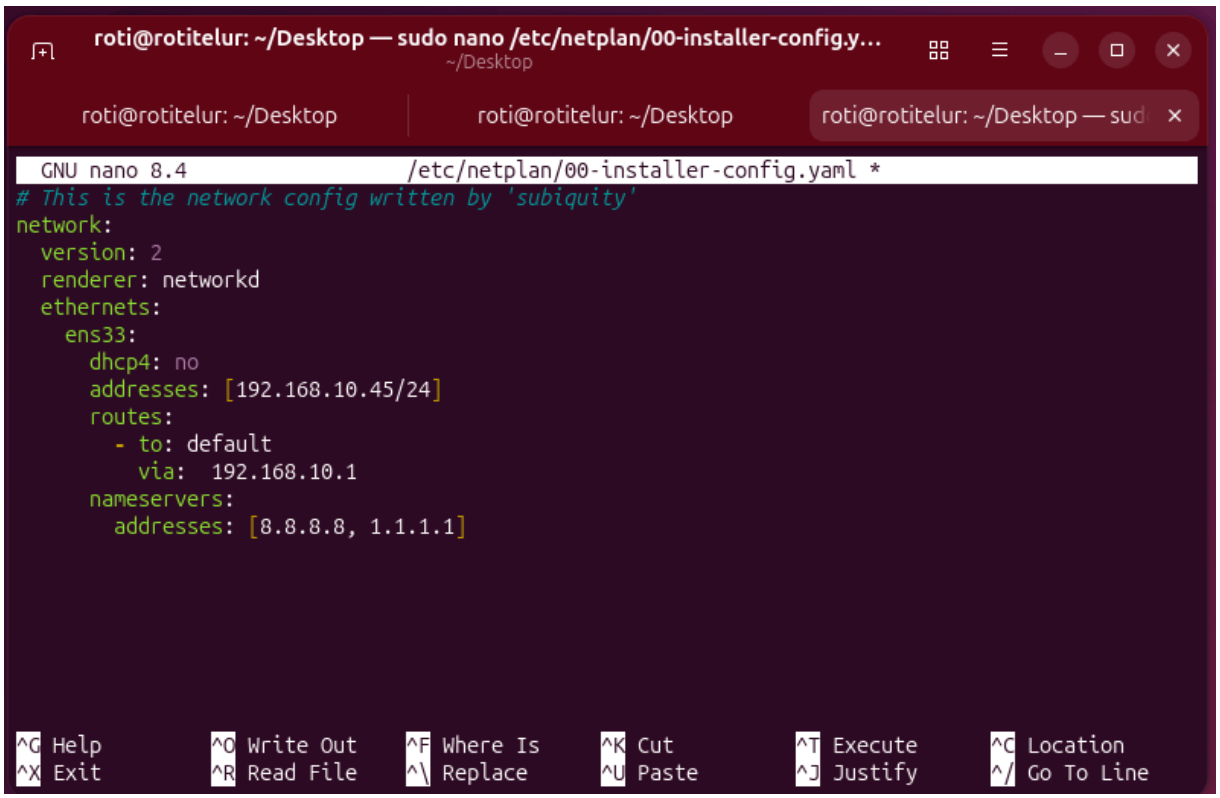
```
Selecting previously unselected package libecap3:amd64.
(Reading database ... 108740 files and directories currently installed.)
Preparing to unpack .../0-libecap3_1.0.1-3.2ubuntu1_amd64.deb ...
Unpacking libecap3:amd64 (1.0.1-3.2ubuntu1) ...
Selecting previously unselected package squid-langpack.
Preparing to unpack .../1-squid-langpack_20191103-1_all.deb ...
Unpacking squid-langpack (20191103-1) ...
Selecting previously unselected package squid-common.
Preparing to unpack .../2-squid-common_4.10-1ubuntu1.13_all.deb ...
Unpacking squid-common (4.10-1ubuntu1.13) ...
Selecting previously unselected package libdbi-perl:amd64.
Preparing to unpack .../3-libdbi-perl_1.643-1ubuntu0.1_amd64.deb ...
Unpacking libdbi-perl:amd64 (1.643-1ubuntu0.1) ...
Selecting previously unselected package ssl-cert.
Preparing to unpack .../4-ssl-cert_1.0.39_all.deb ...
Unpacking ssl-cert (1.0.39) ...
Selecting previously unselected package squid.
Preparing to unpack .../5-squid_4.10-1ubuntu1.13_amd64.deb ...
proxy:x:13:13:proxy:/bin:/usr/sbin/nologin
Unpacking squid (4.10-1ubuntu1.13) ...
Setting up squid-langpack (20191103-1) ...
Setting up ssl-cert (1.0.39) ...
Setting up libdbi-perl:amd64 (1.643-1ubuntu0.1) ...
Setting up libecap3:amd64 (1.0.1-3.2ubuntu1) ...
Setting up squid-common (4.10-1ubuntu1.13) ...
Setting up squid (4.10-1ubuntu1.13) ...
Setcap worked! /usr/lib/squid/pinger is not suid!
Skipping profile in /etc/apparmor.d/disable: usr.sbin.squid
Created symlink /etc/systemd/system/multi-user.target.wants/squid.service → /lib/systemd/system/squi
d.service.
Processing triggers for ufw (0.36-6ubuntu1) ...
Processing triggers for systemd (245.4-4ubuntu3.20) ...
Processing triggers for man-db (2.9.1-1) ...
Processing triggers for libc-bin (2.31-0ubuntu9.9) ...
ubnt4@vm:~$
```


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II) Client Infrastructure Deployment

1. Client-Side Netplan Configuration Script Setup

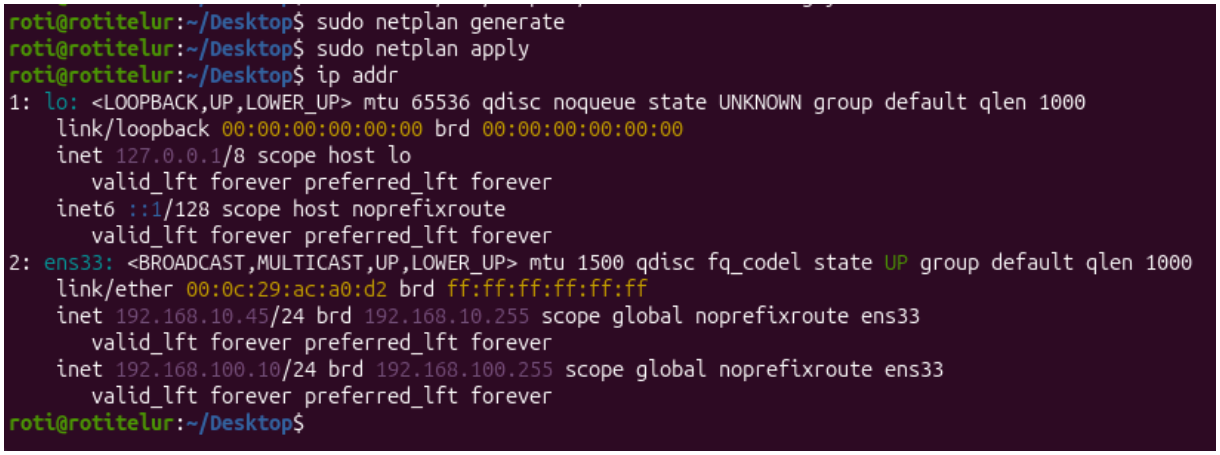
The network configuration file on the Client VM. It assigns the static client IP 192.168.10.45/24 and sets its default route gateway pointing directly to the server at 192.168.10.1.



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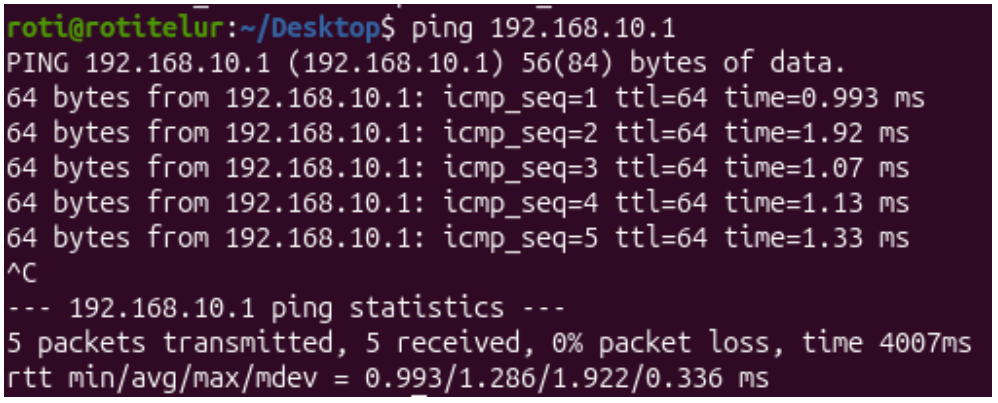
2. Active Client IP Verification Verification Screen

Output logs confirming the client interface ens33 successfully picked up the correct manual static IP addressing mask mapping profile.



3. Gateway Ping Connectivity Test Verification Loop

Execution of a successful ping test from the Client machine to the Ubuntu Server gateway (192.168.10.1), demonstrating 0% packet loss across the host-only virtual switch framework.

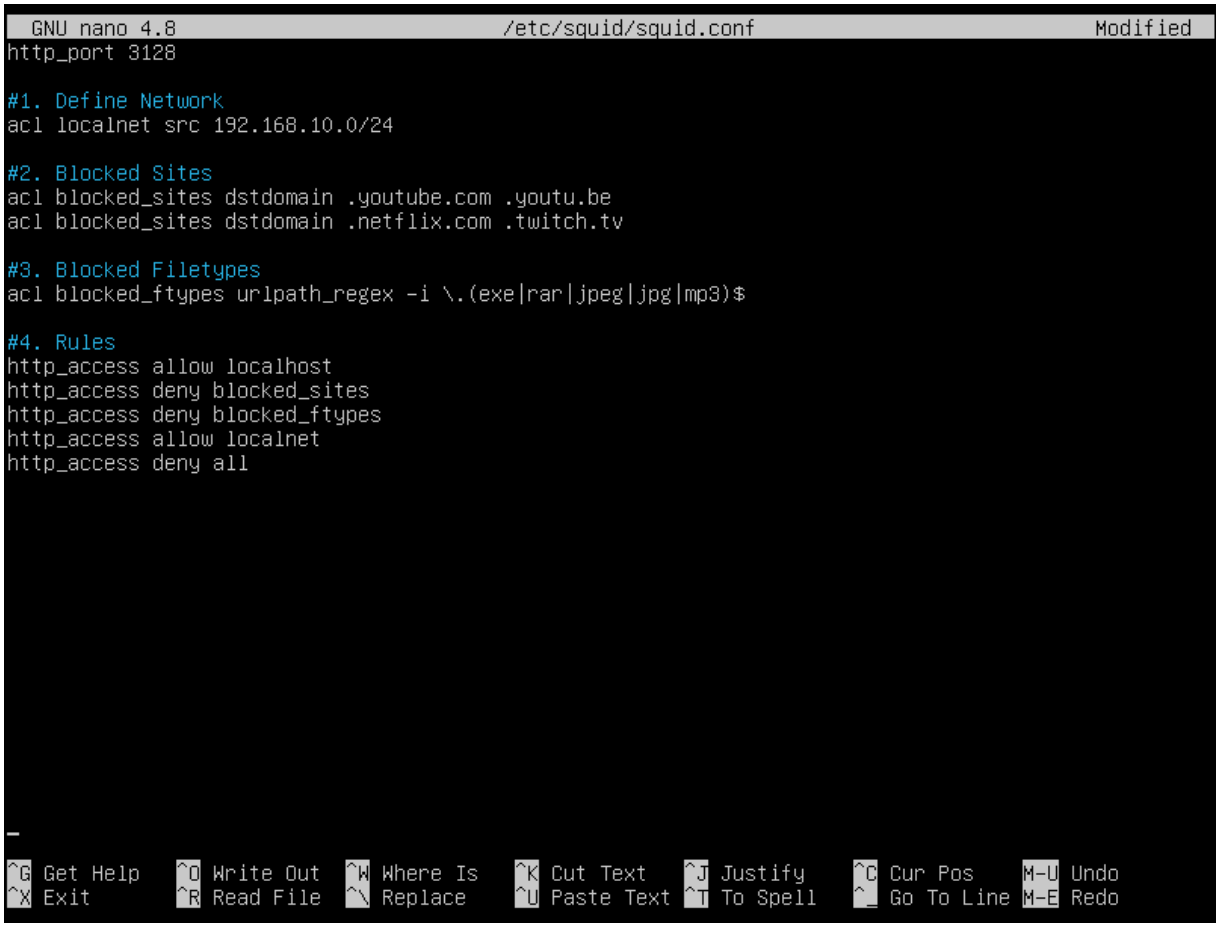


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8.0 SECURITY POLICIES ENFORCEMENT

1. Custom Domain Blocklist Mapping in squid.conf

The Access Control List (ACL) section inside the Squid proxy configuration file. Targeted social media and entertainment domains are restricted via blocked_sites, with the strict enforcement rule ordering placed above general network permissions



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2. Structural Compilation Pass of Squid Rulesets

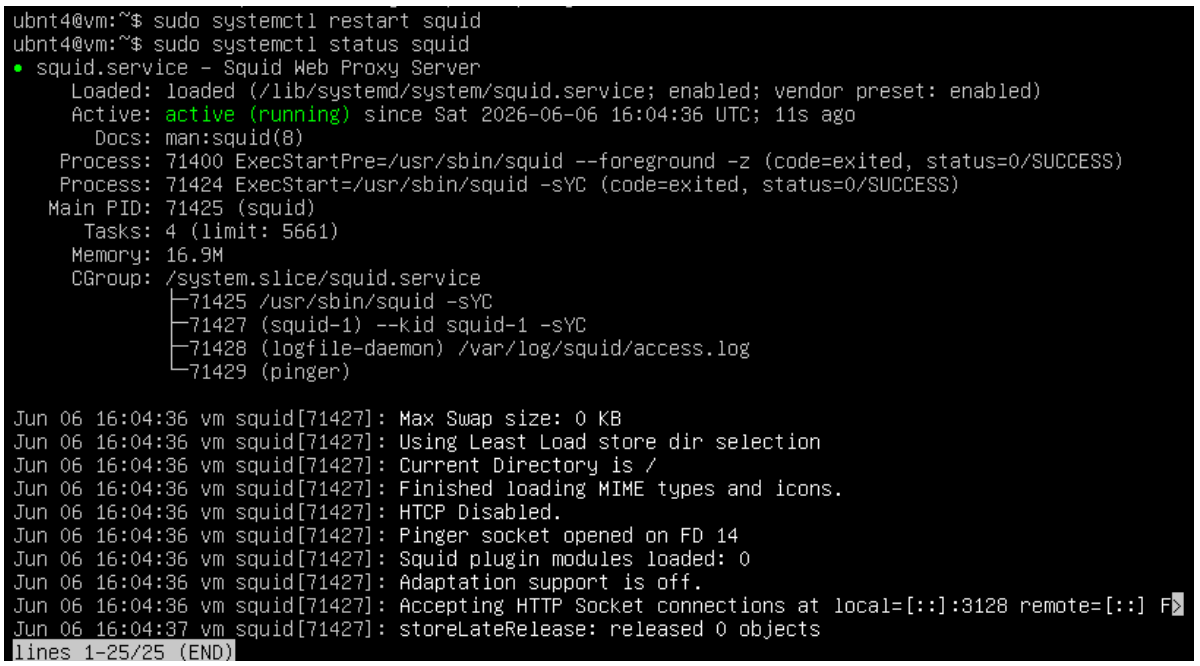
Terminal syntax validation trace executing sudo squid -k parse to guarantee no rule definition anomalies exist before deployment.

```
ubnt4@vm:~$ sudo squid -k reconfigure
ubnt4@vm:~$ sudo squid -k parse
2026/06/07 10:48:14| Startup: Initializing Authentication Schemes ...
2026/06/07 10:48:14| Startup: Initialized Authentication Scheme 'basic'
2026/06/07 10:48:14| Startup: Initialized Authentication Scheme 'digest'
2026/06/07 10:48:14| Startup: Initialized Authentication Scheme 'negotiate'
2026/06/07 10:48:14| Startup: Initialized Authentication Scheme 'ntlm'
2026/06/07 10:48:14| Startup: Initialized Authentication.
2026/06/07 10:48:14| Processing Configuration File: /etc/squid/squid.conf (depth 0)
2026/06/07 10:48:14| Processing: http_port 3128
2026/06/07 10:48:14| Processing: acl localnet src 192.168.10.0/24
2026/06/07 10:48:14| Processing: acl blocked_sites dstdomain .youtube.com .youtu.be
2026/06/07 10:48:14| Processing: acl blocked_sites dstdomain .netflix.com .twitch.tv
2026/06/07 10:48:14| Processing: acl blocked_ftypes urlpath_regex -i \.(exe|rar|jpeg|jpg|mp3)$
2026/06/07 10:48:14| Processing: http_access allow localhost
2026/06/07 10:48:14| Processing: http_access deny blocked_sites
2026/06/07 10:48:14| Processing: http_access deny blocked_ftypes
2026/06/07 10:48:14| Processing: http_access allow localnet
2026/06/07 10:48:14| Processing: http_access deny all
2026/06/07 10:48:14| Initializing https:// proxy context
```


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3. Active Health Verification Tracking of Squid Engine

Active service system control check showing the Squid proxy daemon running normally, listening on the default connection port 3128.



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9.0 NETFILTER FIREWALL DEPLOYMENT

1. Iptables Firewall Configuration Commands Chain

Sequence of netfilter firewall commands applied to clear previous rules, restrict inputs, allow connection port 3128, apply POSTROUTING NAT masquerading, and forward traffic

flush old rules

```
ubnt4@vm:~$ sudo iptables -F
ubnt4@vm:~$ sudo iptables -t nat -F
```

allow basic connections and SSH

```
ubnt4@vm:~$ sudo iptables -A INPUT -i lo -j ACCEPT
ubnt4@vm:~$ sudo iptables -A INPUT -m conntrack --ctstate ESTABLISHED,RELATED -j ACCEPT
ubnt4@vm:~$ sudo iptables -A INPUT -p tcp --dport 22 -j ACCEPT
```

allow the squid proxy

```
ubnt4@vm:~$ sudo iptables -A INPUT -p tcp --dport 3128 -j ACCEPT
```

NAT Masquerading

```
ubnt4@vm:~$ sudo iptables -t nat -A POSTROUTING -o ens33 -j MASQUERADE
```

Forwarding traffic

```
ubnt4@vm:~$ sudo iptables -A FORWARD -i ens35 -o ens33 -j ACCEPT
ubnt4@vm:~$ sudo iptables -A FORWARD -i ens33 -o ens35 -m conntrack --ctstate ESTABLISHED,RELATED -j
 ACCEPT
```

Blocking specific website using IP address

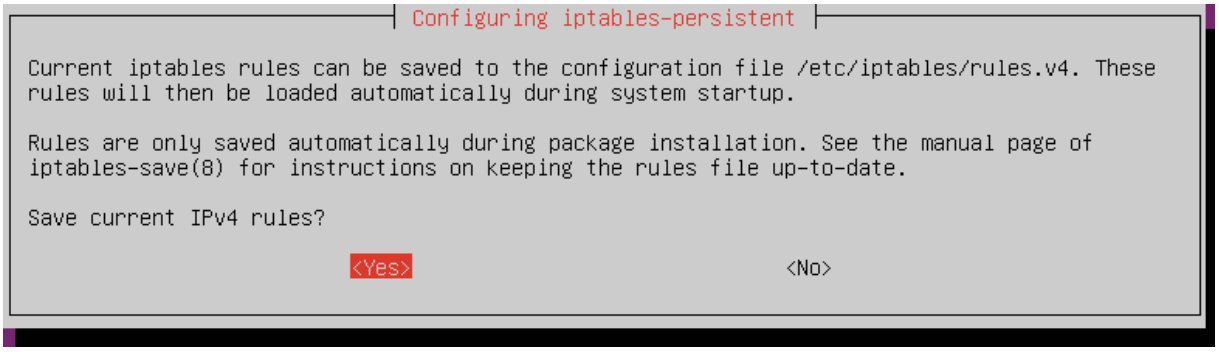
```
ubnt4@vm:~$ sudo iptables -A OUTPUT -d 113.23.169.114 -j DROP
ubnt4@vm:~$
```


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2. IPv4 Netfilter Persistence Mapping Setup Screen

Interactive installation engine save wizard prompted via iptables-persistent to dump active kernel rules securely into standard memory tables

```
ubnt4@vm:~$ sudo apt install iptables-persistent -y
```



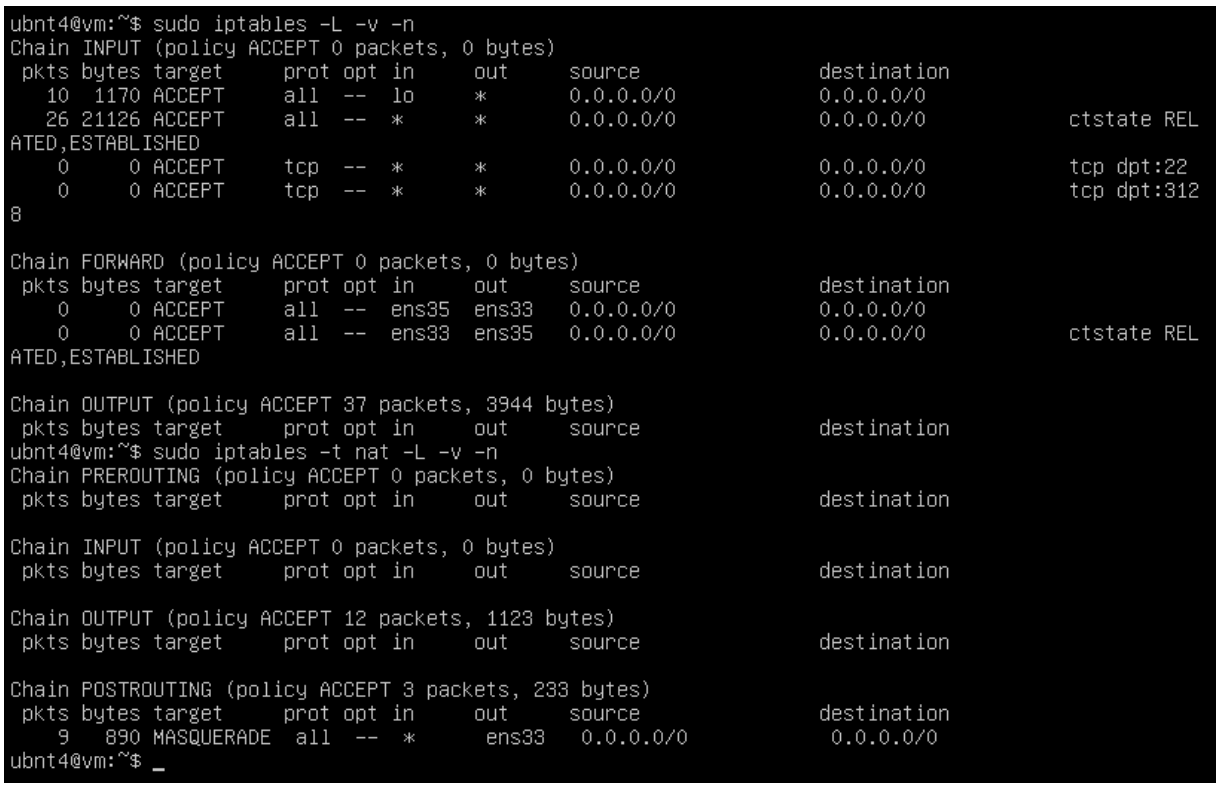

```
Selecting previously unselected package netfilter-persistent.
(Reading database ... 111251 files and directories currently installed.)
Preparing to unpack .../netfilter-persistent_1.0.14ubuntu1_all.deb ...
Unpacking netfilter-persistent (1.0.14ubuntu1) ...
Selecting previously unselected package iptables-persistent.
Preparing to unpack .../iptables-persistent_1.0.14ubuntu1_all.deb ...
Unpacking iptables-persistent (1.0.14ubuntu1) ...
Setting up netfilter-persistent (1.0.14ubuntu1) ...
Created symlink /etc/systemd/system/multi-user.target.wants/netfilter-persistent.service → /lib/systemd/system/netfilter-persistent.service.
Setting up iptables-persistent (1.0.14ubuntu1) ...
update-alternatives: using /lib/systemd/system/netfilter-persistent.service to provide /lib/systemd/system/iptables.service (iptables.service) in auto mode
Processing triggers for man-db (2.9.1-1) ...
Processing triggers for systemd (245.4-4ubuntu3.20) ...
```

```
ubnt4@vm:~$ sudo netfilter-persistent save
run-parts: executing /usr/share/netfilter-persistent/plugins.d/15-ip4tables save
run-parts: executing /usr/share/netfilter-persistent/plugins.d/25-ip6tables save
```


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3. Active Verification Review of Iptables Filters Engine

Full structural evaluation using iptables -L -v -n showing active packet tracking rules, verified byte counts, and operational boundary limits

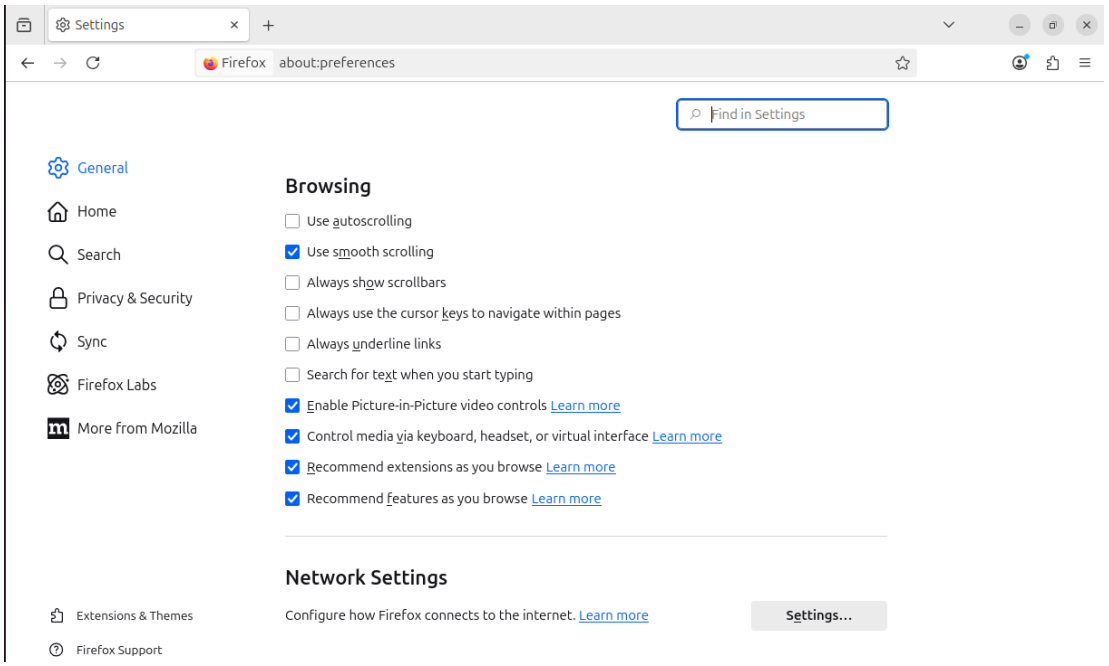


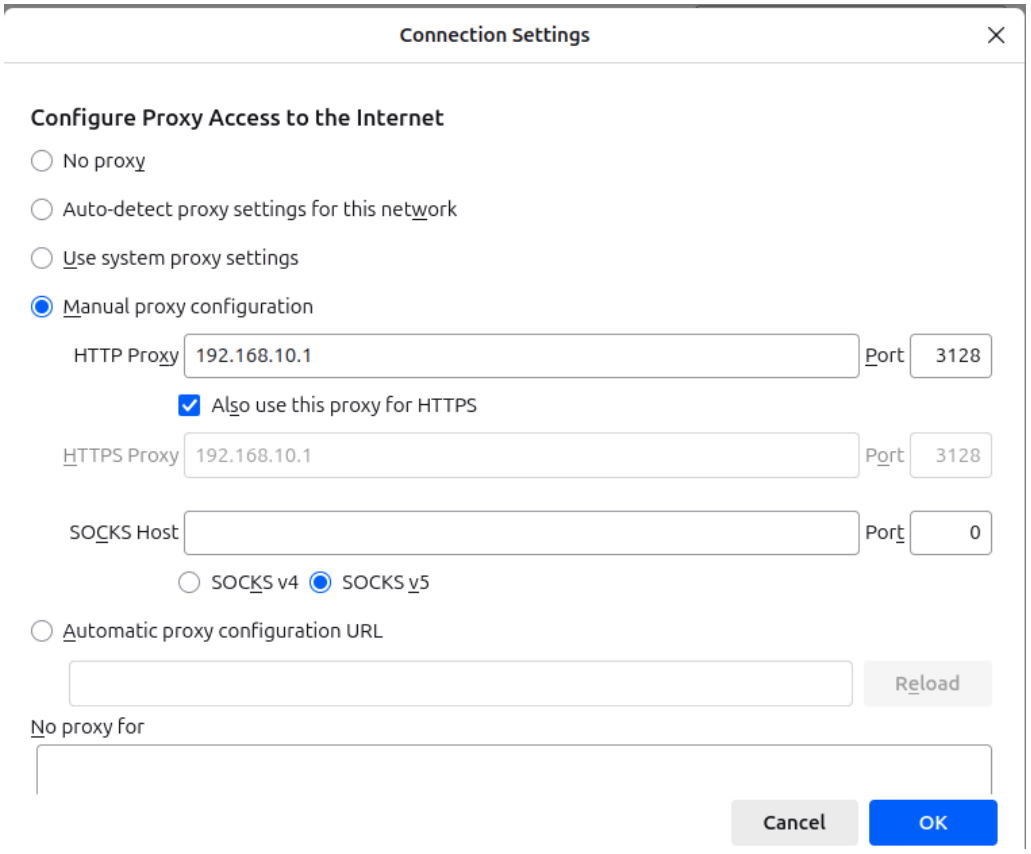
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10.0 TESTING AND VERIFICATION

1. Firefox Browser Manual Proxy Interception Setup

Manual network proxy input options inside Firefox browser mapping traffic redirection rules toward the Ubuntu Server LAN address on communication port 3128.

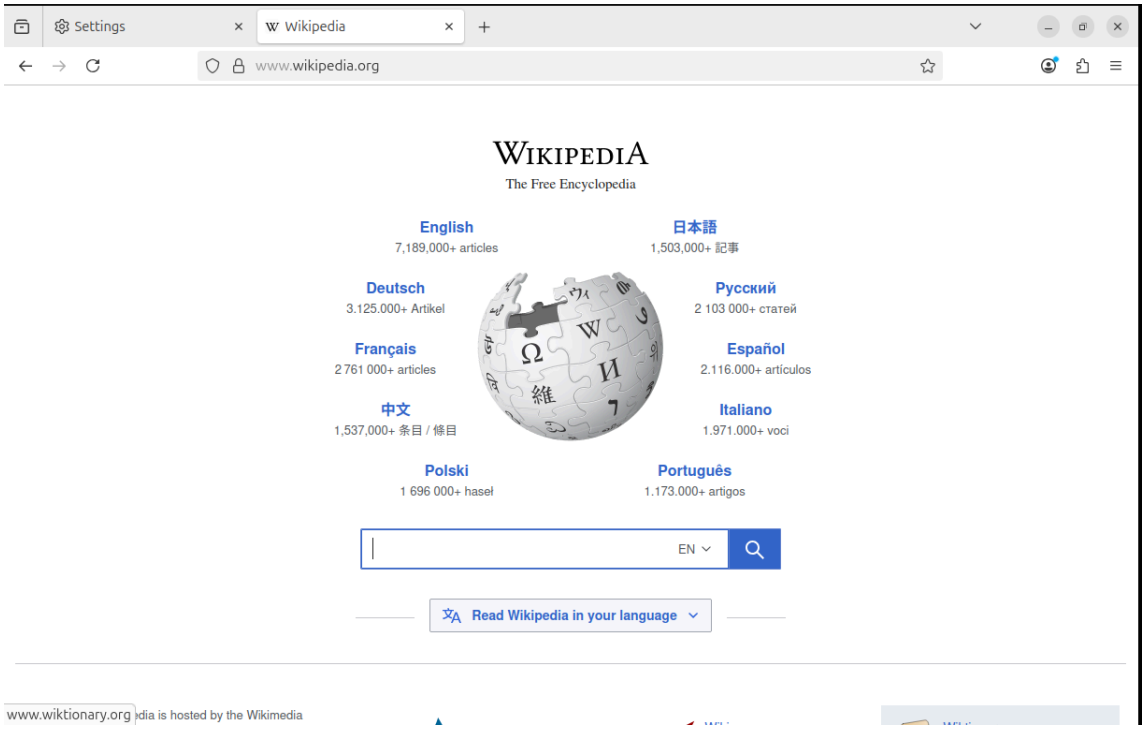




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2. Verification of Unfiltered Content (Wikipedia Whitelist Case)

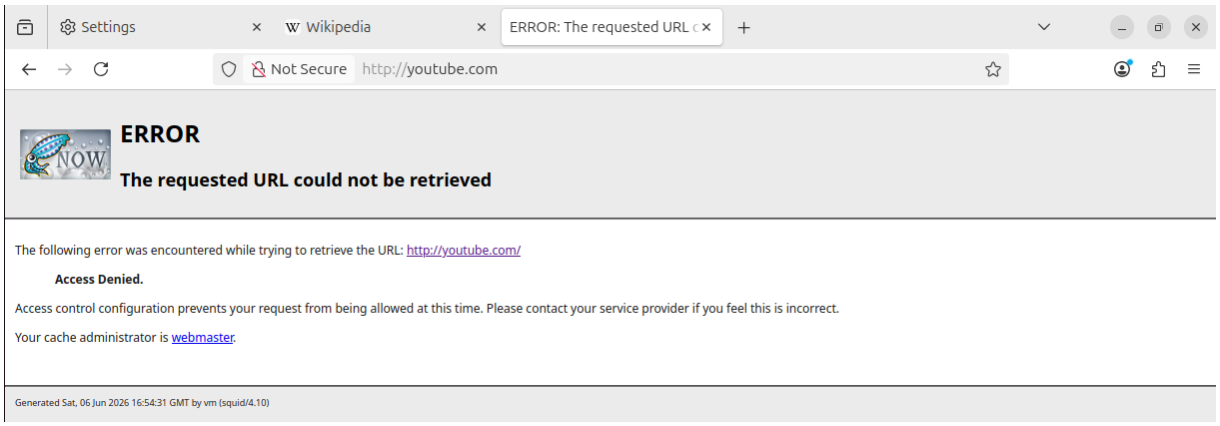
Successful navigation pass loading Wikipedia completely, proving that non-restricted network pathways transmit traffic normally through the gateway.



3. Verification of Application Domain Block

i. youtube.com

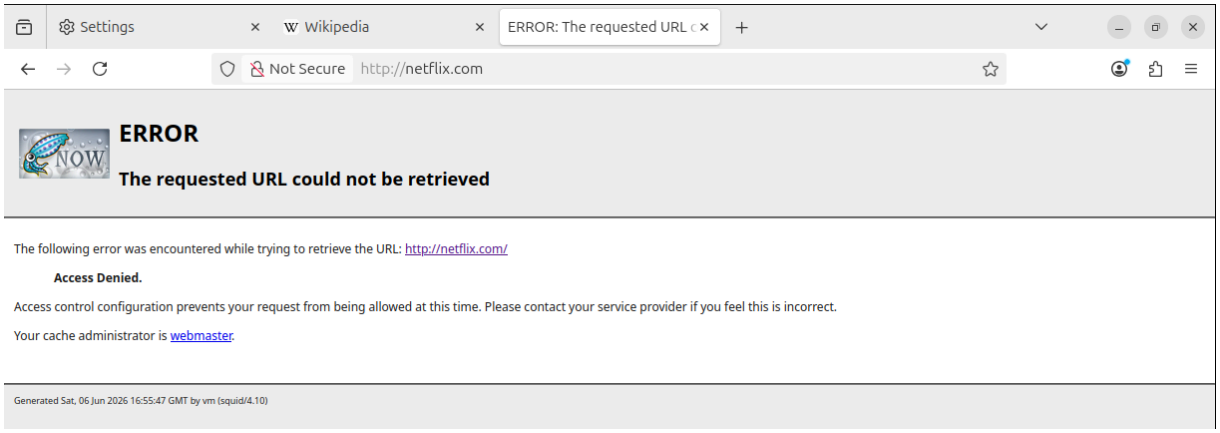
Browser error page was generated directly by the Squid engine blocking connection access to YouTube with a formal Access Denied status response.



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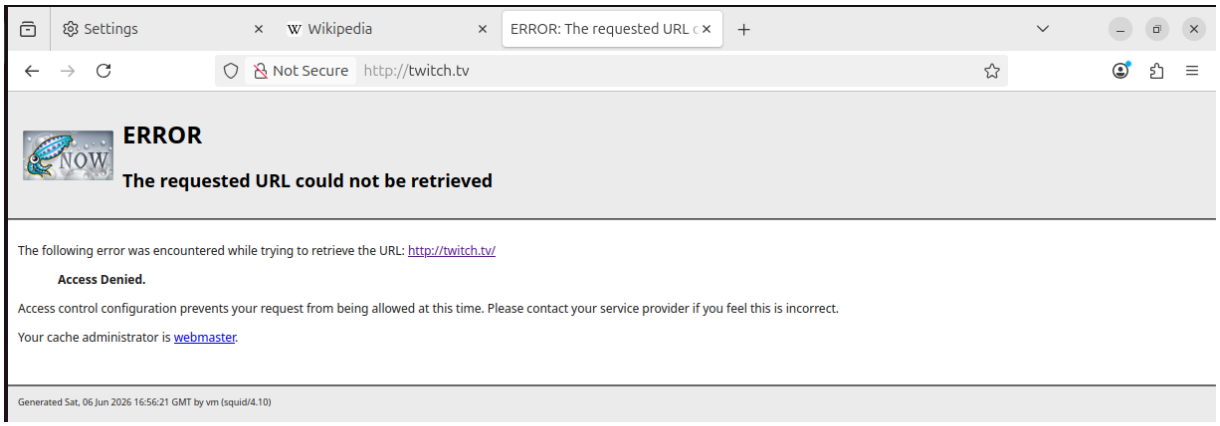
ii. netflix.com

Proxy intersection block tracking screen showing the immediate dropping of connections trying to fetch content from Netflix.



iii. twitch.tv

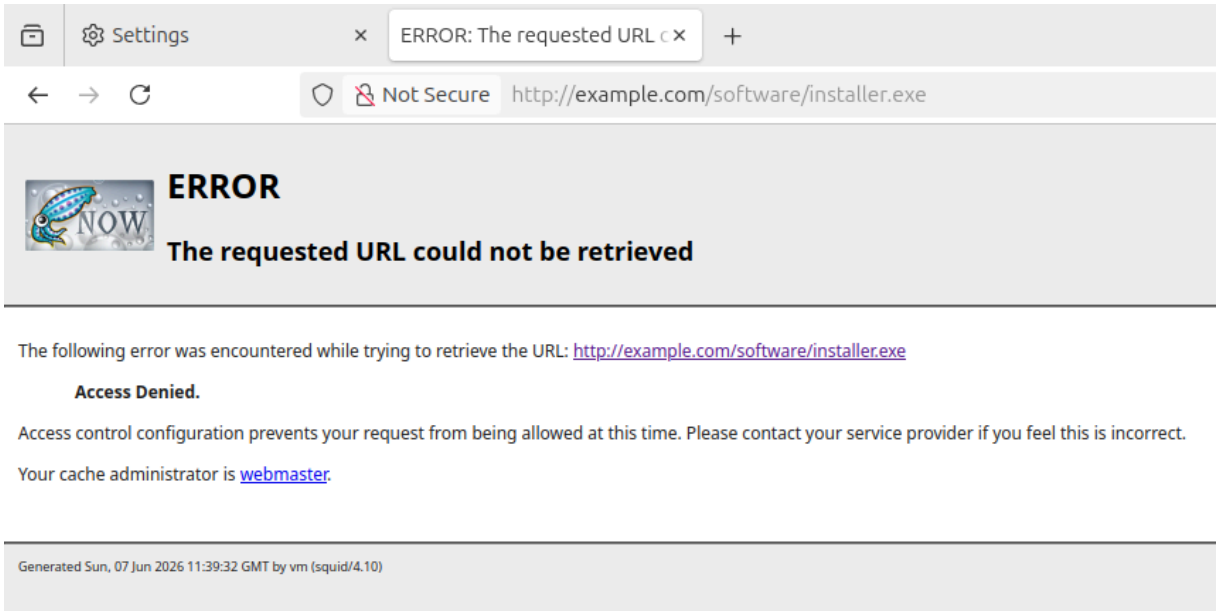
Access block interception validation checking for Twitch entertainment streaming channels, demonstrating absolute compliance enforcement.



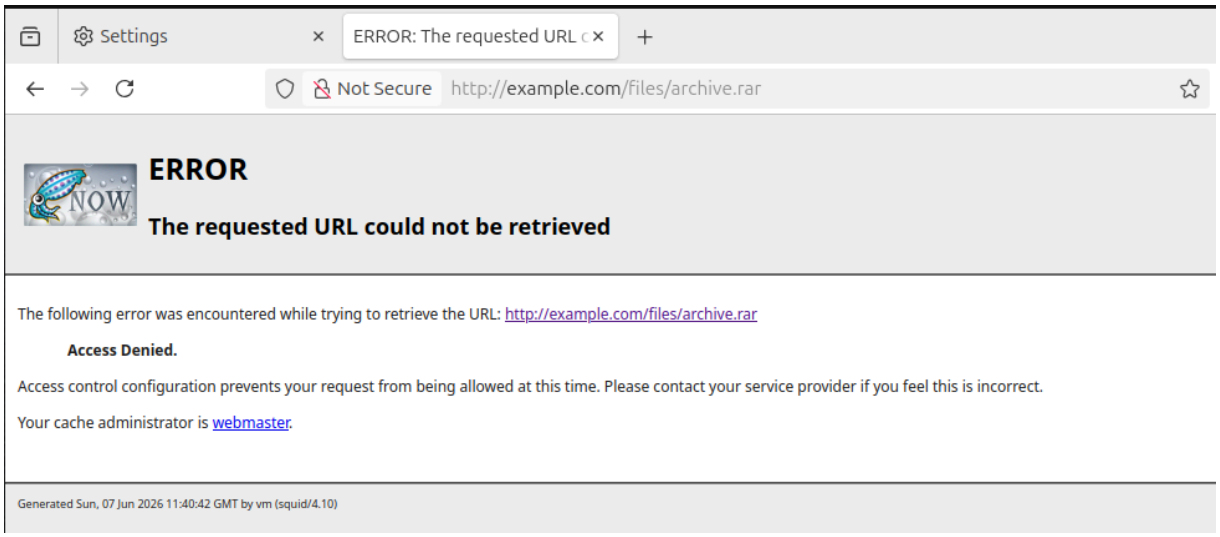
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Verification of Application File Type

i. exe

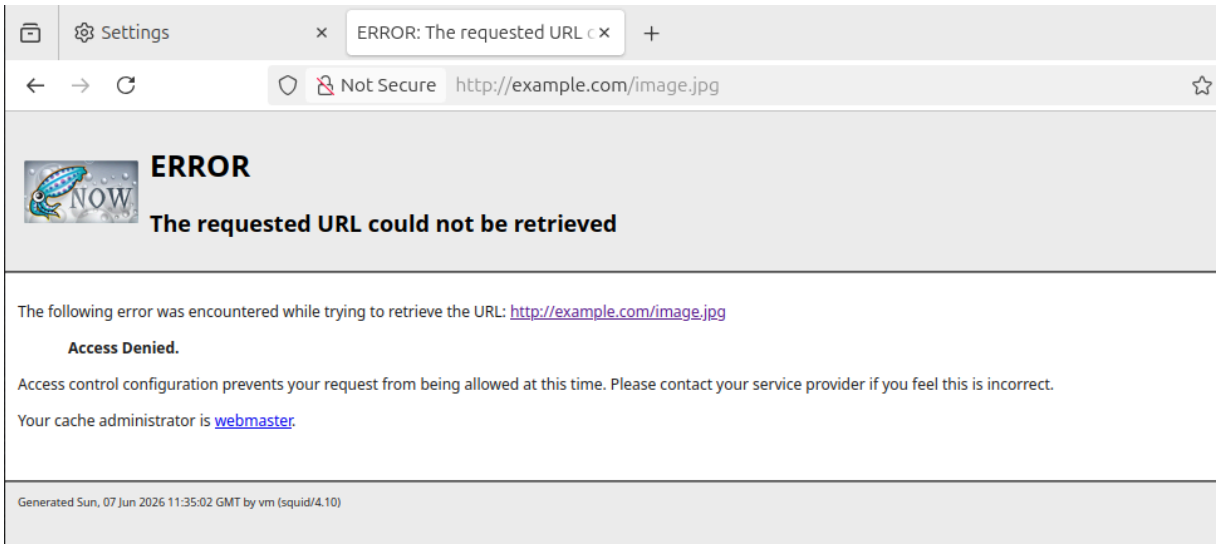


ii. rar

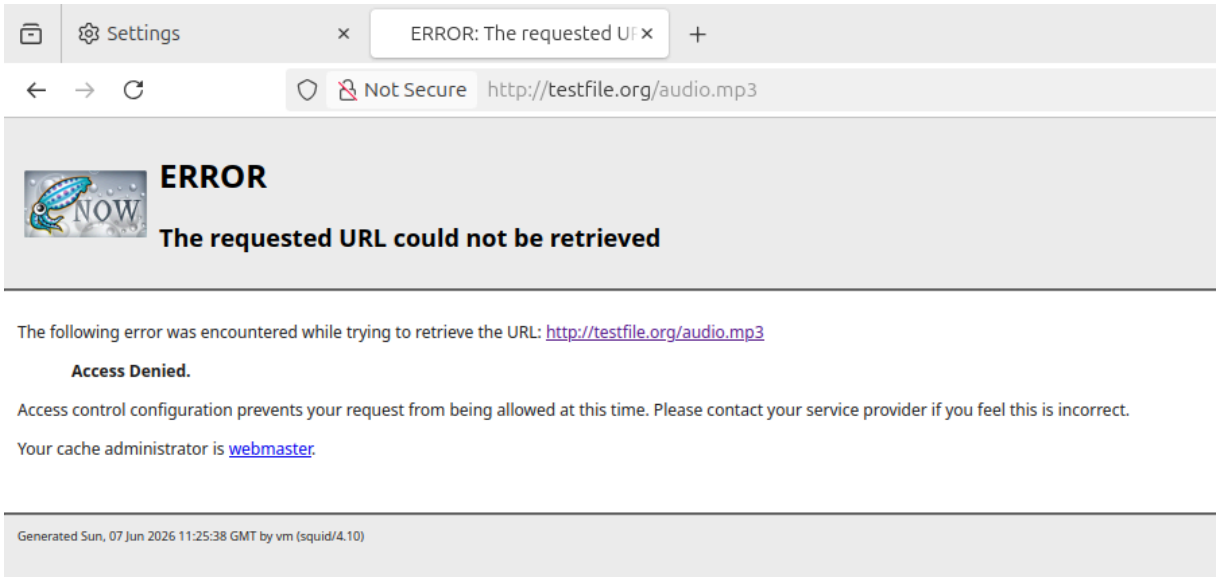


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ii. jpg/jpeg



iv. mp3



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11.0 CONCLUSION

This mini project successfully demonstrated the design, implementation, and testing of a secure virtual network gateway using Ubuntu Server, iptables, and Squid Proxy. The network was built using a virtualized environment where the server was configured with two network adapters. One adapter was used for external internet access through NAT, while the second adapter was used for the internal Host-only LAN connected to the client VM. This setup allowed the Ubuntu Server to act as the main gateway between the internal client network and the external internet network.

The project also achieved its main security objective by applying traffic filtering and access control. iptables was used to control network traffic, enable forwarding, allow required services, and apply NAT masquerading. Squid Proxy was used to filter web access from the client machine. Through Squid access control lists, selected entertainment and social media websites such as YouTube, Netflix, and Twitch were blocked. File type filtering was also implemented to prevent access to selected file extensions such as .exe, .rar, .jpg/.jpeg, and .mp3, as shown in the testing section of the report.

The testing results proved that the configuration worked as expected. The client was able to connect to the Ubuntu Server gateway, normal allowed websites such as Wikipedia could be accessed, while blocked websites and restricted file types were denied by the Squid Proxy. This shows that the security policies were successfully enforced and that the internal client's internet access was controlled through the proxy server.

Overall, this project helped improve understanding of virtual network design, Linux network configuration, IP addressing, firewall rules, proxy filtering, and security testing. It also showed how open-source tools can be used to create a simple but effective internet control system for a small office or lab environment. Although HTTPS filtering has limitations without SSL Bump, the project still met the main objective of demonstrating basic traffic filtering, website blocking, file type blocking, and secure gateway implementation.

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